

# EXPERIMENT: PRIME PALINDROME IN JAVA

|                |                                                                         |
|----------------|-------------------------------------------------------------------------|
| Experiment No. | 02                                                                      |
| Date           | 17/09/2026                                                              |
| Title          | Java Program to Check Whether a Given Number is Prime Palindrome or Not |

## Objective / Aim

To write and execute a Java program to check whether the given number is both a prime number and a palindrome.

## System Requirements

### Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM

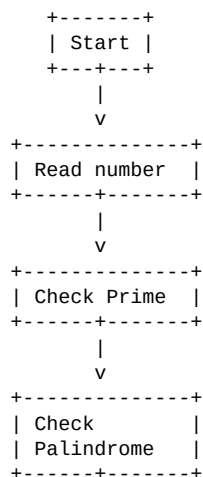
### Software

- Windows / Linux / macOS
- Java Development Kit (JDK 17 or later)
- Text Editor or Java IDE
- Command Prompt / Terminal

## Algorithm / Logic Description

1. Start the program.
2. Read a number from the user.
3. Check whether the number is prime by testing divisibility from 2 up to  $n/2$ .
4. Reverse the number using the modulus (%) and division (/) operators.
5. Compare the reversed number with the original number to check whether it is a palindrome.
6. If the number is both prime and palindrome, display that it is a Prime Palindrome.
7. Otherwise, display that it is not a Prime Palindrome.
8. Stop the program.

## Flow / Logic Diagram



```

      |
      v
+-----+
| Prime AND Palindrome?|
+-----+
      | Yes      | No
      v          v
+-----+ +-----+
| Prime      | | Not Prime |
| Palindrome | | Palindrome |
+-----+ +-----+
      |          |
      +-----+
      v
      +-----+
      | End |
      +-----+

```

## Source Code

```

import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Check prime
        boolean prime = true;

        if (n <= 1) {
            prime = false;
        } else {
            for (int i = 2; i <= n / 2; i++) {
                if (n % i == 0) {
                    prime = false;
                    break;
                }
            }
        }

        // Check palindrome
        int temp = n;
        int reverse = 0;

        while (temp > 0) {
            int digit = temp % 10;
            reverse = reverse * 10 + digit;
            temp = temp / 10;
        }

        // Final result
        if (prime && n == reverse) {
            System.out.println(n + " is a Prime Palindrome.");
        } else {
            System.out.println(n + " is not a Prime Palindrome.");
        }

        sc.close();
    }
}

```

## Compilation & Execution Commands

```

javac PrimePalindrome.java
java PrimePalindrome

```

## Sample Output

```

Enter a number: 131
131 is a Prime Palindrome.

```

## Result

The Java program was successfully implemented and executed to check whether a given number is a Prime Palindrome. For the sample input 131, the program correctly identifies it as a Prime Palindrome.